

Docker Networking Lab Handbook

BE (.NET Web API) to PostgreSQL Connection - From Scratch

Learning Objectives

- Understand Docker networking
- Create Docker networks
- Run PostgreSQL and .NET containers
- Connect API to PostgreSQL using container names
- Troubleshoot networking issues

Architecture

- Customer API Container -> PostgreSQL Container
- Containers communicate through a Docker network

Step 1 - Create Network

- `docker network create training-network`

Step 2 - Run PostgreSQL

- `docker run -d --name postgres-db --network training-network -e POSTGRES_USER=postgres -e POSTGRES_PASSWORD=postgres -e POSTGRES_DB=customerdb postgres:16`

Step 3 - Create .NET Web API

- `dotnet new webapi -n CustomerApi`

Step 4 - Configure Connection String

- Host=postgres-db;Port=5432;Database=customerdb;Username=postgres;Password=postgres

Step 5 - Build Docker Image

- `docker build -t customer-api:1.0 .`

Step 6 - Run API Container

- `docker run -d --name customer-api --network training-network -p 8080:8080 customer-api:1.0`

Verification

- `docker network inspect training-network`
- `docker logs customer-api`

Golden Rule

- Inside Docker, do NOT use localhost for another container.
- Use the container name: postgres-db